Buying Different Tech Stocks, or Buying the Same Risk Repeatedly?

A quantitative investigation of diversification, statistical commonality, downside risk and portfolio construction

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Historical sample: January 2020 – September 2026

Executive summary

Eight technology-related equities—AAPL, MSFT, NVDA, META, GOOGL, AMD, TSM and ASML—were examined using aligned daily adjusted-price returns. Their average pairwise correlation was 0.549, and the first principal component (PC1) explained 60.67% of standardized-return variance over the full sample. Equal weighting reduced volatility relative to the average constituent volatility but did not balance risk contributions: AMD and NVDA jointly accounted for 35.01% of portfolio volatility contribution with just 25% of capital.

Common movement was stronger in the selected COVID-19 and 2022 bear-market windows than in 2023 but was not stable through time: the PC1 share declined to 47.85% in 2024–Sep 2026. Out-of-sample minimum-variance optimization reduced annualized volatility by 20.4% versus equal weights, without eliminating market exposure. Thus, diversification among related stocks is real but incomplete and time dependent.

Research design

Design element	Specification
Universe	AAPL, MSFT, NVDA, META, GOOGL, AMD, TSM, ASML
Data	Yahoo Finance adjusted daily closes; common trading dates; returns in USD
Full sample	2020-01-02 to 2026-09-30; 1,694 daily return observations
Training / testing	2020–2023 (1,005) / 2024–Sep 2026 (689)
Portfolio rules	Long-only, fully invested; up to 35% per stock; constant target weights
Strategies	Equal Weight; Minimum Variance; historical Minimum CVaR at 95%
Benchmarks	SPY and XLK; separate single-benchmark regressions

All figures are historical exploratory/backtest outputs, not forecasts. Reported returns are hypothetical and exclude commissions, taxes, spreads and market impact. The stock universe was chosen retrospectively.

Computational Framework

The analysis was conducted in Python using pandas and NumPy for financial data processing, scikit-learn for Principal Component Analysis (PCA), SciPy for constrained portfolio optimization, and Matplotlib for data visualization. Historically adjusted closing prices were retrieved from Yahoo Finance through the yfinance library.

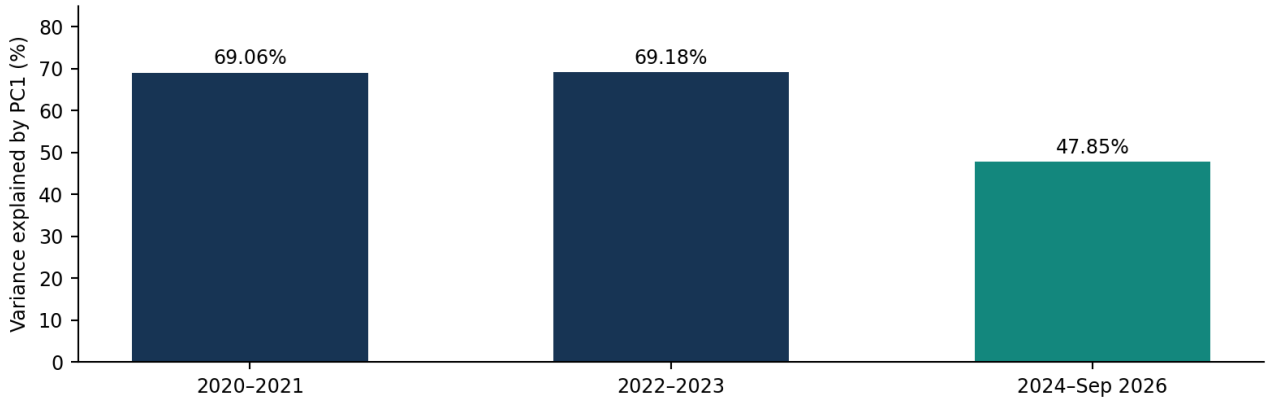
Portfolio optimization was implemented using quadratic programming for Minimum Variance and linear programming based on the Rockafellar–Uryasev formulation for Minimum CVaR at the 95% confidence level. Model performance was evaluated using an out-of-sample framework to assess portfolio risk and return beyond the optimization period.

1 Diversification and PCA

Measure	Observed result
Average pairwise daily-return correlation	0.549
Mean constituent annualized volatility	41.10%
Equal-weight annualized volatility	32.17%
Relative reduction vs. mean constituent volatility	21.7%
Full-sample PC1 explained variance	60.67%
Effective PCA participation dimension	2.53

The 21.7% figure compares equal-weight volatility to the arithmetic average of individual volatilities—not to SPY or to a portfolio constructed outside technology. PCA was fitted to standardized daily returns (equivalent to using their correlation structure); its components reflect statistical co-movement, not automatically identifiable economic causes.

1.1 Stability across subsamples

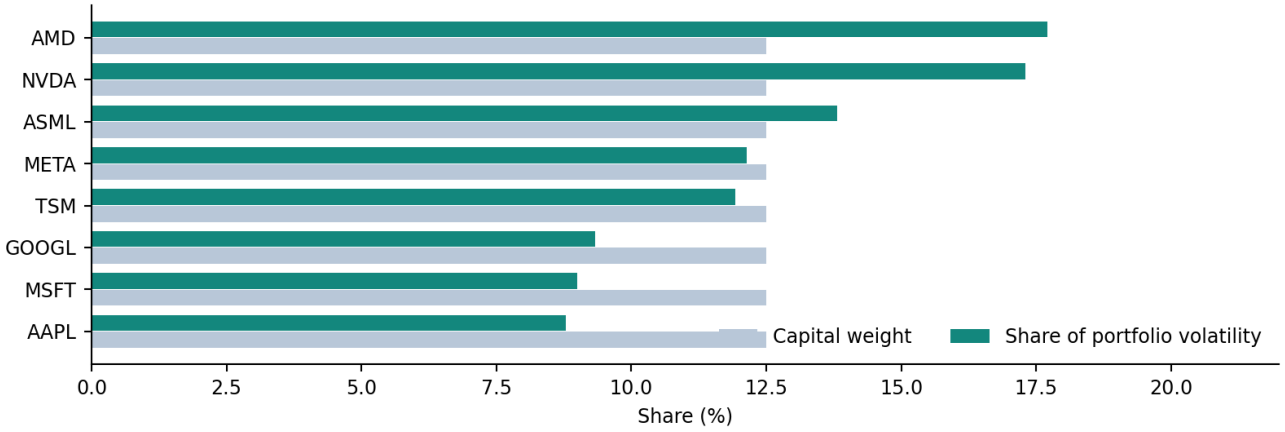


Period	PC1 explained	PC1+PC2 explained	Effective dimension
2020–2021	69.06%	78.58%	2.02
2022–2023	69.18%	78.43%	2.02
2024–Sep 2026	47.85%	61.65%	3.64

PC1 became less dominant in the latest subperiod. The participation dimension is an eigenvalue-concentration index, not a literal count of independent economic bets. Differences across periods may reflect changes in correlation structure and need not persist.

2 Capital allocation is not risk allocation

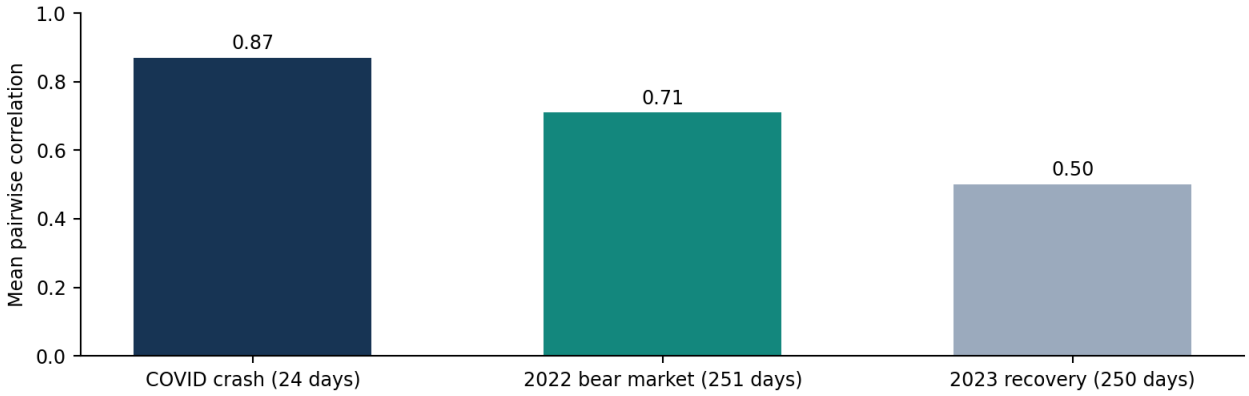
The equal-weight portfolio allocates 12.5% to each stock. Its Euler decomposition of volatility using the annualized covariance matrix yields different contributions to total volatility. A larger share indicates more influence on portfolio volatility, accounting for both individual volatility and covariance with other positions.



Capital weight versus percentage contribution to portfolio volatility; Jan 2020–Sep 2026.

AMD (17.71%) and NVDA (17.30%) jointly contributed 35.01% of the volatility attribution, even though they represented only 25% of invested capital. The same calculation gives AAPL and MSFT a combined 17.79% risk share.

3 Historical stress-period co-movement



Selected window	Daily observations	Mean corr.	Annualized portfolio vol.
COVID crash, 2020	24	0.87	88.17%
Bear market, 2022	251	0.71	42.18%
Recovery, 2023	250	0.50	25.91%

These windows were selected after observing historical events and have unequal lengths; particularly, the COVID estimate is based on only 24 observations. The comparisons support higher realized common movement in these chosen stress windows, not the universal claim that correlations necessarily rise whenever markets fall.

4 Portfolio optimization and out-of-sample evaluation

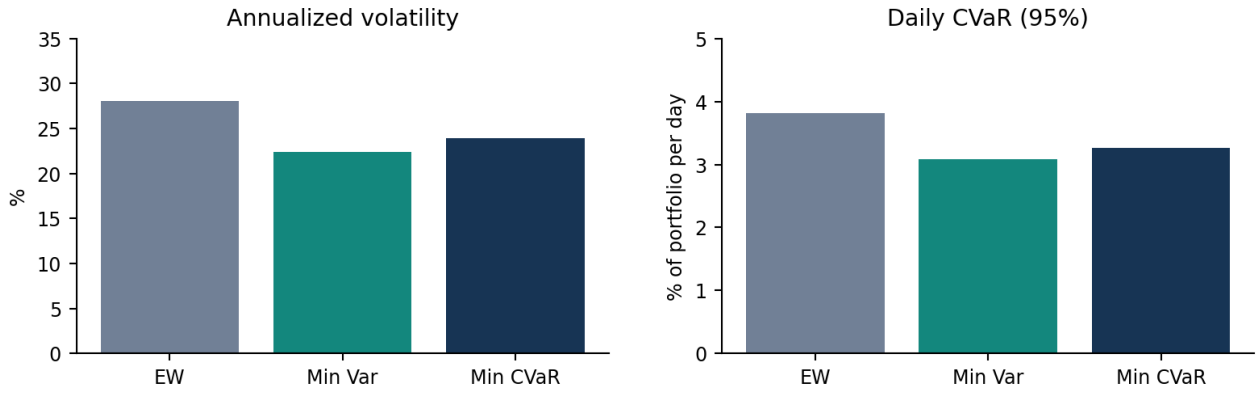
Minimum Variance minimized portfolio variance using the 2020–2023 covariance matrix. Minimum CVaR minimized empirical expected shortfall of daily portfolio losses at 95% confidence through linear programming. Each strategy was long-only, fully invested and subject to a 35% position limit. No common return target was imposed, so this is a comparison of risk-minimization objectives, not equal-expected-return portfolios.

Stock	Equal Weight	Minimum Variance	Minimum CVaR
AAPL	12.50%	24.33%	18.81%
MSFT	12.50%	25.85%	26.48%
NVDA	12.50%	0.00%	0.00%
META	12.50%	0.00%	0.00%
GOOGL	12.50%	26.68%	19.72%
AMD	12.50%	0.00%	0.00%
TSM	12.50%	23.14%	35.00%
ASML	12.50%	0.00%	0.00%

The optimized zero weights are consequences of the objectives, bounds and training period; they do not show that these excluded securities are intrinsically poor investments.

4.1 Out-of-sample performance (2024–Sep 2026)

Strategy	Annual return	Annual volatility	Daily CVaR 95%	Max drawdown
Equal Weight	50.01%	28.08%	3.81%	−28.99%
Minimum Variance	38.23%	22.36%	3.08%	−26.79%
Minimum CVaR	42.87%	23.96%	3.27%	−27.70%



Separate scales and horizons: annualized standard deviation (left) and daily 95% CVaR (right).

Minimum Variance reduced annualized volatility by 20.4% and daily empirical CVaR by 19.2% relative to Equal Weight. Although Minimum CVaR minimized tail risk in training, Minimum Variance had the smaller tail-loss estimate in the held-out period. Equal Weight earned more while exhibiting greater measured risk.

5 Market and sector benchmark exposure

Portfolio	SPY beta	SPY R^2	XLK beta	XLK R^2
Equal Weight	1.556	74.6%	0.991	83.3%
Minimum Variance	1.210	71.0%	0.708	66.9%
Minimum CVaR	1.279	69.1%	0.786	71.9%

All three portfolios remained positively related to both benchmarks. Betas across SPY and XLK are not interchangeable units, because benchmark volatilities differ. Separate single-benchmark regressions do not identify the incremental technology factor after controlling the broad market; moreover, XLK does not fully represent the selected multinational universe.

6 Rebalancing sensitivity

Portfolio	Daily ann. return	Monthly ann. return	Daily vol.	Monthly vol.
Equal Weight	50.01%	49.99%	28.08%	28.13%
Minimum Variance	38.23%	37.88%	22.36%	22.37%
Minimum CVaR	42.87%	42.47%	23.96%	24.03%

Monthly target-weight rebalancing produced similar strategy rankings to the hypothetical daily rebalanced portfolios in the selected test window. This does not establish robustness to actual trading costs or to changes in training dates.

7 Interpretation and limitations

The empirical evidence favors a nuanced answer: owning more technology stocks does not necessarily produce proportionately more independent sources of risk. The eight-stock basket received a diversification benefit, yet exhibited co-movement, uneven volatility contributions and substantial positive benchmark exposure. Changing portfolio weights mitigated volatility but did not eliminate exposure to common fluctuations.

Key limitations: selection of large successful survivors with knowledge of the historical period; a single train/test partition; static optimized target weights; omission of implementation costs and taxes; relatively short and unequally sized event windows; no confidence intervals; and sector exposure assessed through separate single-factor regressions. Claims about broader technology universes or future markets therefore remain provisional.